

Problem 1

Given the following twist ξ , what is the homogeneous transformation matrix?

(a) $\xi = [v^\top \ \Omega^\top]^\top = [3/\pi \ 0 \ 0 \ \pi/3 \ 0 \ 0]^\top.$

(b) $\xi = [v^\top \ \Omega^\top]^\top = [0 \ 0.5 \ 0 \ 0 \ \pi/2 \ 0]^\top.$

(c) $\xi = [v^\top \ \Omega^\top]^\top = [1 \ 0 \ 0 \ 0 \ 0 \ \pi/3]^\top.$

Part (a)

$$T = \begin{bmatrix} 1.0000 & 0 & 0 & 1.0000 \\ 0 & 0.5000 & -0.8660 & 0 \\ 0 & 0.8660 & 0.5000 & 0 \\ 0 & 0 & 0 & 1.0000 \end{bmatrix}$$

Part (b)

$$T = \begin{bmatrix} 0 & 0 & 1.0000 & 0 \\ 0 & 1.0000 & 0 & 0.7854 \\ -1.0000 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1.0000 \end{bmatrix}$$

Part (c)

$$T = \begin{bmatrix} 0.5000 & -0.8660 & 0 & 0.8660 \\ 0.8660 & 0.5000 & 0 & 0.5000 \\ 0 & 0 & 1.0000 & 0 \\ 0 & 0 & 0 & 1.0000 \end{bmatrix}$$

Problem 2

Given the homogeneous transformation matrix,

$$\mathbf{T} = \begin{bmatrix} 0.5000 & -0.6124 & 0.6124 & 0.8415 \\ 0.6124 & 0.7500 & 0.2500 & 0.3251 \\ -0.6124 & 0.2500 & 0.7500 & -0.3251 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- (a) What is the displacement?
- (b) What is the Quaternion that encodes the rotation?
- (c) What is the Angle-Axis that encodes the rotation?
- (d) What is the twist ξ that encodes the full transform?

Part (a): Displacement

The displacement is the upper-right 3×1 block of $\mathbf{T}$:

$$\mathbf{d} = \begin{bmatrix} 0.8415 \\ 0.3251 \\ -0.3251 \end{bmatrix}$$

Part (b): Quaternion

With $\text{tr}(\mathbf{R}) = 0.5 + 0.75 + 0.75 = 2$:

$$\begin{aligned} q_0 &= \sqrt{\frac{1 + \text{tr}(\mathbf{R})}{4}} = \sqrt{\frac{3}{4}} = 0.866 \\ q_1 &= \frac{r_{32} - r_{23}}{4q_0} = \frac{0.25 - 0.25}{4(0.866)} = 0 \\ q_2 &= \frac{r_{13} - r_{31}}{4q_0} = \frac{0.6124 - (-0.6124)}{4(0.866)} = 0.354 \\ q_3 &= \frac{r_{21} - r_{12}}{4q_0} = \frac{0.6124 - (-0.6124)}{4(0.866)} = 0.354 \end{aligned}$$

Check: $q_0^2 + q_1^2 + q_2^2 + q_3^2 = 0.75 + 0 + 0.125 + 0.125 = 1$.

$$\mathbf{Q} = 0.866 + \begin{bmatrix} 0 \\ 0.354 \\ 0.354 \end{bmatrix}$$

Part (c): Angle-Axis

The vector of off-diagonal differences is

$$\begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix} = \begin{bmatrix} 0 \\ 1.2248 \\ 1.2248 \end{bmatrix}, \quad \|\cdot\|_2 = \sqrt{0^2 + 1.2248^2 + 1.2248^2} = 1.7321.$$

Then

$$\theta = \text{atan2}\left(\frac{1}{2} \|\cdot\|_2, \frac{\text{tr}(\mathbf{R}) - 1}{2}\right) = \text{atan2}(0.866, 0.5) = \frac{\pi}{3} \text{ rad}$$

$$\hat{\mathbf{k}} = \frac{1}{2 \sin \theta} \begin{bmatrix} 0 \\ 1.2248 \\ 1.2248 \end{bmatrix} = \frac{1}{1.7321} \begin{bmatrix} 0 \\ 1.2248 \\ 1.2248 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.707 \\ 0.707 \end{bmatrix}$$

$$\mathbf{\Omega} = \theta \hat{\mathbf{k}} = \frac{\pi}{3} \begin{bmatrix} 0 \\ 0.707 \\ 0.707 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.741 \\ 0.741 \end{bmatrix}$$

Part (d): Twist

The angular part is $\mathbf{\Omega}$ from Part (c). The linear part follows the course's inverse-twist formula (with $\boldsymbol{\xi} = [\mathbf{v}; \theta \hat{\mathbf{k}}]$, where $\mathbf{v}$ is the linear part per radian):

$$\mathbf{v} = \left(\frac{\sin \theta}{2(1 - \cos \theta)} \mathbf{I} + \frac{2(1 - \cos \theta) - \theta \sin \theta}{2\theta(1 - \cos \theta)} \hat{\mathbf{k}} \hat{\mathbf{k}}^\top - \frac{1}{2} [\hat{\mathbf{k}}]_\times \right) \mathbf{d}.$$

For $\theta = \pi/3$: $\frac{\sin \theta}{2(1 - \cos \theta)} = 0.866$, so

$$\mathbf{v} = \left(0.866 \mathbf{I} + 0.0889 \hat{\mathbf{k}} \hat{\mathbf{k}}^\top - \frac{1}{2} \begin{bmatrix} 0 & -0.707 & 0.707 \\ 0.707 & 0 & 0 \\ -0.707 & 0 & 0 \end{bmatrix} \right) \begin{bmatrix} 0.8415 \\ 0.3251 \\ -0.3251 \end{bmatrix}.$$

Since $\hat{\mathbf{k}}^\top \mathbf{d} = 0.707(0.3251) + 0.707(-0.3251) = 0$, the $\hat{\mathbf{k}} \hat{\mathbf{k}}^\top$ term drops out, and

$$\mathbf{v} = 0.866 \begin{bmatrix} 0.8415 \\ 0.3251 \\ -0.3251 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} -0.4597 \\ 0.5950 \\ -0.5950 \end{bmatrix} = \begin{bmatrix} 0.9586 \\ -0.016 \\ 0.016 \end{bmatrix}.$$

$$\boldsymbol{\xi} = [\mathbf{v}^\top \quad \mathbf{\Omega}^\top]^\top = [0.959 \quad -0.016 \quad 0.016 \quad 0 \quad 0.741 \quad 0.741]^\top$$

Problem 3

For the following give one pro and one con for using it as a representation,

- (a) Rotation and displacement separately.
- (b) Homogeneous transformation matrices.
- (c) Twist coordinates.

Part (a): Rotation and Displacement Separately

Pro: Easy to identify what is happening.

Con: Rotation and displacement are separate objects.

Part (b): Homogeneous Transformation Matrices

Pro: Composition is a single matrix multiply, so chaining frames is easy.

Con: Homogeneous transforms are non-minimal (16 numbers for 6 DOF) and require enforcing orthogonality constraints on R under numerical operations (e.g. repeated composition/interpolation).

Part (c): Twist Coordinates

Pro: Provides velocity of object.

Con: It is singular or ambiguous at $\theta = 0$ and $\theta = \pi$.

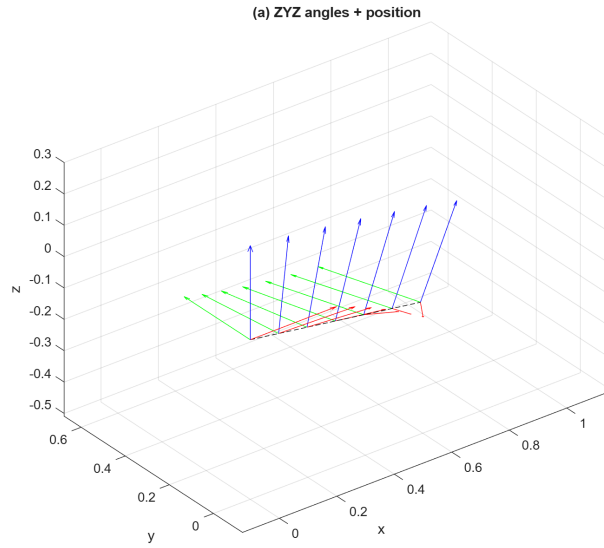
Problem 4

Given the following two homogeneous transformations, use MATLAB to plot a linear interpolation between the two. Your plot should contain a minimum of 5 intermediate points and should have the orthogonal axis (columns of the rotation matrix) drawn at each point. *Hint*: Use the `quiver3` function.

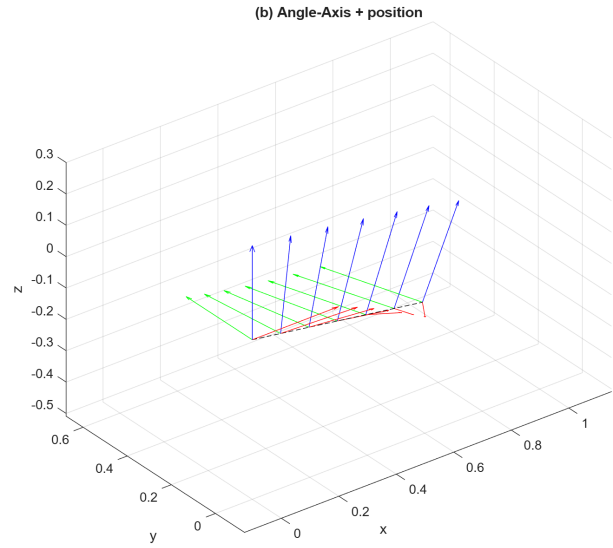
$$\mathbf{T}_{\text{initial}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{T}_{\text{final}} = \begin{bmatrix} 0.5000 & -0.6124 & 0.6124 & 0.8415 \\ 0.6124 & 0.7500 & 0.2500 & 0.3251 \\ -0.6124 & 0.2500 & 0.7500 & -0.3251 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- (a) Perform the interpolation using ZYZ angles and position.
- (b) Perform the interpolation using Angle-Axis and position.
- (c) Perform the interpolation using Quaternion and position (remember to enforce the length constraint on the quaternion before calculating the intermediate orientations).
- (d) Perform the interpolation using twist.
- (e) Comment on the similarities or differences.

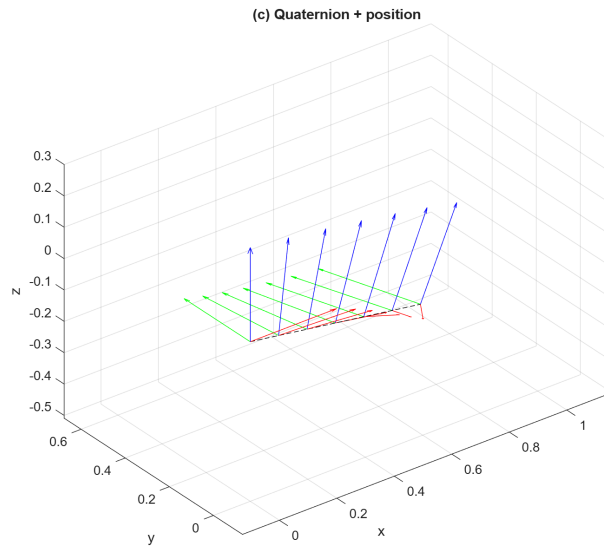
Parts (a)–(d): Interpolation Results



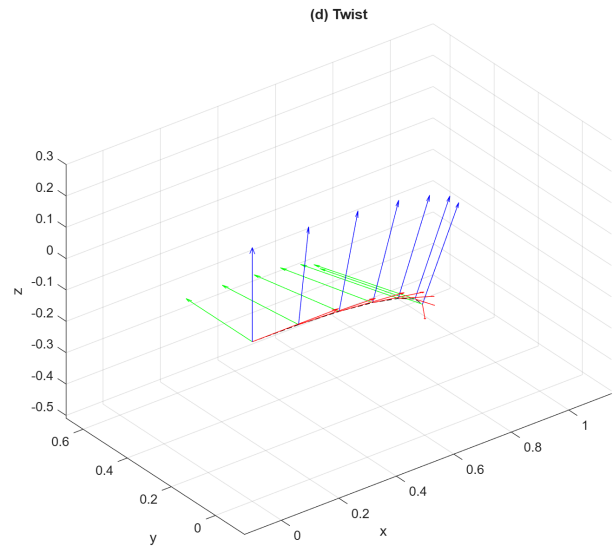
(a) ZYZ-angle and linear-position interpolation.



(b) Angle-axis and linear-position interpolation.



(c) Normalized-quaternion and linear-position interpolation.



(d) Twist interpolation.

Part (e): Comparison

All four start at the identity and end at T_{final} , so the only difference is what happens in between.

For orientation, angle-axis, quaternion, and twist all rotate about the same fixed axis $\hat{k} = [0, 0.707, 0.707]^T$ through a total of $\pi/3$, so they match. The quaternion is linearly interpolated and renormalized, so its speed isn't perfectly constant, but for only 60° you can't tell. ZYZ interpolates the three angles separately, so it isn't a single-axis rotation.

For position, ZYZ, angle-axis, and quaternion use linear interpolation, so the origin moves in a straight line. Twist does rotation and translation together as a screw motion, so the origin follows a curved (helical) path instead.

Appendix A: Code

Problem 1

Listing 1: Problem1.m

```
1 root = fileparts(fileparts(fileparts(fileparts(mfilename('fullpath')))));
2 addpath(fullfile(root, 'Functions'));
3
4
5 % Twists are  $\xi = [v; \omega]$ 
6 xi_a = [3/pi; 0; 0; pi/3; 0; 0];
7 xi_b = [0; 0.5; 0; 0; pi/2; 0];
8 xi_c = [1; 0; 0; 0; 0; pi/3];
9
10 T_a = twist2Transform(xi_a);
11 T_b = twist2Transform(xi_b);
12 T_c = twist2Transform(xi_c);
13
14 disp('(a) T ='); disp(T_a);
15 disp('(b) T ='); disp(T_b);
16 disp('(c) T ='); disp(T_c);
```

Problem 4

Listing 2: Problem4.m

```

1 root = fileparts(fileparts(fileparts(fileparts(mfilename('fullpath')))));
2 addpath(fullfile(root, 'Functions'));
3 figDir = fullfile(fileparts(mfilename('fullpath')), '..', 'Latex', '
    Figures');
4 if ~exist(figDir, 'dir'), mkdir(figDir); end
5
6 T_i = eye(4);
7 T_f = [ 0.5000 -0.6124  0.6124  0.8415;
8         0.6124  0.7500  0.2500  0.3251;
9        -0.6124  0.2500  0.7500 -0.3251;
10         0        0        0        1 ];
11
12 R_i = T_i(1:3,1:3);  p_i = T_i(1:3,4);
13 R_f = T_f(1:3,1:3);  p_f = T_f(1:3,4);
14
15 N = 5;                                     % intermediate points (minimum 5)
16 s = linspace(0, 1, N + 2);               % includes both endpoints
17
18 [phi, theta, psi] = rot2ZYZ(R_i);  zyz_i = [phi(1); theta(1); psi(1)];
19 [phi, theta, psi] = rot2ZYZ(R_f);  zyz_f = [phi(1); theta(1); psi(1)];
20 Ra = cell(1, numel(s));  pa = zeros(3, numel(s));
21 for k = 1:numel(s)
22     zyz = zyz_i + s(k) * (zyz_f - zyz_i);
23     Ra{k} = rotZ(zyz(1)) * rotY(zyz(2)) * rotZ(zyz(3));
24     pa(:,k) = p_i + s(k) * (p_f - p_i);
25 end
26 plotFrames(Ra, pa, '(a) ZYZ angles + position', fullfile(figDir, 'p4a.png'
    ));
27
28
29 om_i = rot2AngleAxis(R_i);
30 om_f = rot2AngleAxis(R_f);
31
32 Rb = cell(1, numel(s));  pb = zeros(3, numel(s));
33 for k = 1:numel(s)
34     om = om_i + s(k) * (om_f - om_i);
35     Rb{k} = angleAxis2Rot(om); % TODO
36     pb(:,k) = p_i + s(k) * (p_f - p_i);
37 end
38 plotFrames(Rb, pb, '(b) Angle-Axis + position', fullfile(figDir, 'p4b.png'
    ));
39
40 Rc = cell(1, numel(s));  pc = zeros(3, numel(s));
41 Q_i = rot2Quat(R_i);
42 Q_f = rot2Quat(R_f);
43
44 if dot(Q_i, Q_f) < 0
45     Q_f = -Q_f;
46 end
47

```

```

48 for k = 1:numel(s)
49     Q = Q_i + s(k) * (Q_f-Q_i);
50     Q = Q / norm(Q);
51     Rc{k} = quat2Rot(Q);
52     pc(:,k) = p_i + s(k) * (p_f - p_i);
53 end
54 plotFrames(Rc, pc, '(c) Quaternion + position', fullfile(figDir, 'p4c.png'
    ));
55
56 x_i = transform2Twist(T_f);
57 Rd = cell(1, numel(s)); pd = zeros(3, numel(s));
58 for k = 1:numel(s)
59     x = [x_i(1:3) ; s(k)* x_i(4:6)];
60     if s(k) == 0
61         T = T_i;      % twist2Transform treats a zero-rotation twist as pure
                        % translation by v
62     else
63         T = twist2Transform(x);
64     end
65     Rd{k} = T(1:3,1:3);
66     pd(:,k) = T(1:3, 4);
67 end
68 plotFrames(Rd, pd, '(d) Twist', fullfile(figDir, 'p4d.png'));
69
70
71
72 function plotFrames(Rs, ps, ttl, file)
73     figure; hold on; grid on; axis equal;
74     plot3(ps(1,:), ps(2,:), ps(3,:), 'k--');
75     for k = 1:numel(Rs)
76         R = Rs{k}; p = ps(:,k);
77         quiver3(p(1), p(2), p(3), R(1,1), R(2,1), R(3,1), 0.3, 'r');
78         quiver3(p(1), p(2), p(3), R(1,2), R(2,2), R(3,2), 0.3, 'g');
79         quiver3(p(1), p(2), p(3), R(1,3), R(2,3), R(3,3), 0.3, 'b');
80     end
81     xlabel('x'); ylabel('y'); zlabel('z'); title(ttl);
82     view(3); hold off;
83     exportgraphics(gcf, file, 'Resolution', 200);
84 end

```

Appendix B: Hand Calculations

Problem 2

$$T = \begin{bmatrix} .5 & -.6124 & .6124 & .8415 \\ .6124 & .75 & .25 & .3251 \\ -.6124 & .25 & .75 & -.3251 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

a) $\underline{d} = \begin{bmatrix} .8415 \\ .3251 \\ -.3251 \end{bmatrix}$

b) $a_0 = \pm \sqrt{\frac{1 + \overset{2}{\text{Tr}(R)}}{4}} = .866$

$$a_1 = \frac{r_{32} - r_{23}}{4a_0} = \frac{.25 - .25}{4 \cdot .866} = 0$$

$$a_2 = \frac{r_{13} - r_{31}}{4a_0} = \frac{.6124 - (-.6124)}{4 \cdot .866} = .354$$

$$a_3 = \frac{r_{21} - r_{12}}{4a_0} = \frac{.6124 - (-.6124)}{4 \cdot .866} = .354$$

$$Q = .866 + \begin{bmatrix} 0 \\ .354 \\ .354 \end{bmatrix}$$

$$C) \quad \theta = \arctan 2 \left(\frac{1}{2} \left\| \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix} \right\|_2, \frac{\pm r(R) - 1}{2} \right)$$

$$\begin{array}{rcl} 0 & & .25 - .25 \\ 1.2248 & \leftarrow & .6124 - (-.6124) \\ 1.2248 & & .6124 - (-.6124) \end{array}$$

$$\sqrt{0^2 + 1.225^2 + 1.225^2} = 1.73241$$

$$\theta = \arctan 2(.866, .5)$$

$$\theta = \pi/3 \text{ rad}$$

$$\hat{K} = \frac{1}{2.5\theta} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix} = \frac{1}{2.5\theta} \begin{bmatrix} 0 \\ 1.225 \\ 1.225 \end{bmatrix}$$

$$\hat{K} = \begin{bmatrix} 0 \\ .637 \\ .637 \end{bmatrix}$$

$$\underline{\underline{\hat{K}}} = \frac{\pi}{3} \cdot \begin{bmatrix} 0 \\ .707 \\ .707 \end{bmatrix}$$

$$d) \underline{v} = \left(\frac{s\theta}{2(1-\cos\theta)} \underline{I} + \frac{2(1-\cos\theta) - \theta s\theta}{2\theta(1-\cos\theta)} \hat{k} \hat{k}^T - \frac{1}{2} [\hat{k}]_{\times} \right) \underline{d}$$

$$= \left(\frac{s \frac{\pi}{3}}{2(1-\cos \frac{\pi}{3})} \underline{I} + \frac{2(1-\cos \frac{\pi}{3}) - \frac{\pi}{3} s \frac{\pi}{3}}{2 \frac{\pi}{3} (1-\cos \frac{\pi}{3})} \begin{bmatrix} 0 \\ .707 \\ .707 \end{bmatrix} \begin{bmatrix} 0 \\ .707 \\ .707 \end{bmatrix}^T - \frac{1}{2} \begin{bmatrix} 0 & -.707 & .707 \\ .707 & 0 & 0 \\ -.707 & 0 & 0 \end{bmatrix} \right) \cdot \begin{bmatrix} .8415 \\ .3251 \\ -.3251 \end{bmatrix}$$

$$\underline{v} = [.9586, -.016, .016]^T$$

$$\underline{\omega} = \underline{\Omega} = \theta \cdot \hat{k} = \begin{bmatrix} 0 \\ .74 \\ .74 \end{bmatrix}$$

$$\underline{\xi} = \begin{bmatrix} .959 \\ -.016 \\ .016 \\ 0 \\ .74 \\ .74 \end{bmatrix}$$

Remainder of problems were
coded / written in latex